G-CAM

Version 6.6

Raspberry Pi AI Camera Platform

System Documentation

RealTech Systems

Confidential — Internal Documentation

Document Summary

Document	G-Cam V6.6 System Documentation
Platform	Raspberry Pi (ARM64 / ARMv7)
Version	6.6
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This document is the official technical reference for G-Cam Version 6.6, an AI-powered surveillance camera platform built on Raspberry Pi. It covers system architecture, installation procedures, configuration, detection logic, MQTT command reference, and operational guidelines.

1. System Overview

G-Cam is an AI-powered surveillance camera system that runs on a Raspberry Pi. It connects to an IP camera over RTSP, performs real-time detection of people, vehicles, and garbage, and reports events through MQTT messages, SFTP file uploads, and a local web dashboard.

1.1 Core Capabilities

- Real-time person, vehicle, and garbage detection using AI models
- RTSP video stream ingestion and MJPEG web streaming
- MQTT-based command and event reporting
- SFTP upload of snapshots, video clips, and audio
- Local Flask web dashboard with geofence editors and live feed
- Cloudflare tunnel integration for secure remote access
- Audio warning system with live push-to-talk
- Battery monitoring via INA226 over I2C
- Scheduled and on-demand power saving mode
- GPIO-driven router reboot capability

1.2 Technology Stack

Component	Technology
AI Detection	MobileNetSSD (person/vehicle) + YOLOv8 ONNX (garbage)
Web Framework	Flask + Waitress WSGI server
Messaging	MQTT via paho-mqtt
File Transfer	SFTP via paramiko
Remote Access	Cloudflare Tunnel (cloudflared)
Audio	espeak-ng + FFmpeg + aplay / ALSA
Database / State	JSON files (config.json + state.json)
Service Management	systemd services and timers
Power Monitoring	INA226 I2C sensor
GPIO	RPi.GPIO (router reboot)

2. Installation

The installer script (gcam_full_reinstall_v6.6_fixed.sh) fully configures the system from scratch. It must be run with sudo privileges and executes 17 sequential steps.

2.1 Pre-Installation Checklist

☐	Copy garbage_yolo.onnx to the Pi user's home directory (e.g., /home/pi/garbage_yolo.onnx)
☐	Confirm the RTSP URL is correct and accessible from the Pi
☐	Confirm MQTT broker is reachable from the Pi
☐	Confirm SFTP server is reachable and all remote paths exist (or user has create permission)
☐	Run the installer as root: sudo bash gcam_full_reinstall_v6.6_fixed.sh
☐	After install, verify: sudo systemctl status gcam.service shows active (running)

2.2 Installation Steps

STEP **01**

Stop Old Services

Kills previously running G-Cam processes and removes old service files so the system starts clean.

STEP **02**

Remove Old Application

Deletes the entire /opt/gcam folder so nothing from a previous install carries over.

STEP **03**

Fix DNS if Needed

Checks internet reachability. If unreachable, replaces DNS config with Google (8.8.8.8) and Cloudflare (1.1.1.1).

STEP **04**

Update apt Package List

Runs apt-get update to refresh the package list from all configured repositories.

STEP **05**

Install System Packages

Installs all required Linux packages: Python, FFmpeg, ALSA tools, OpenCV dependencies, I2C tools, GPIO tools, espeak-ng, curl, and more.

STEP

Install Cloudflared

06 Downloads and installs the Cloudflare tunnel binary (cloudflared) for the correct CPU architecture (ARM64 or ARMv7).

STEP **07**

Configure ALSA Audio

Sets the default sound device to the USB speaker at hardware slot hw:0,0 by writing /etc/asound.conf.

STEP **08**

Create Directory Structure

Creates the complete folder hierarchy under /opt/gcam/ including assets, data, logs, models, files, and templates.

STEP **09**

Create Python Virtual Environment

Creates a Python venv at /opt/gcam/venv and installs all required libraries: Flask, OpenCV, paho-mqtt, paramiko, psutil, smbus2, and waitress.

STEP **10**

Download Detection Models

Downloads MobileNetSSD person/vehicle model files (.prototxt + .caffemodel) from GitHub. Copies garbage_yolo.onnx from the Pi home directory.

STEP **11**

Create Warning Audio

Uses espeak-ng to generate a spoken warning message, then processes it through FFmpeg to reduce noise and normalize volume.

STEP **12**

Write config.json

Saves all values entered during setup into /opt/gcam/data/config.json — the single source of truth for the application.

STEP **13**

Write Helper Scripts

Creates battery voltage reader, safe reboot script, and sudoers rules allowing the Pi user to manage gcam.service without a password.

STEP **14**

Write HTML Templates

Creates three web pages: main dashboard (index.html), full-screen live feed (live.html), and push-to-talk page (live_talk.html).

STEP **15**

Write app.py

Writes the main application (~5,000+ lines). Handles RTSP capture, AI detection, Flask web server, MQTT listener, SFTP uploads, audio, and Cloudflare tunnels.

STEP 16

Write controller.py

Writes a companion process that handles MQTT commands requiring control of the main gcam.service: Close/Open Software and Camera Live URL.

STEP 17

Write Services & Timers, Start Everything

Creates and enables all systemd units, then starts the full application stack.

3. Configuration

The installer prompts for configuration values interactively. All values are saved to /opt/gcam/data/config.json and serve as the single source of truth for the application.

3.1 Configuration Parameters

Parameter	Example Value	Description
Device ID	RTGCAMx	Unique name used in filenames and MQTT topics
Camera Name	RTGCAMx	Human-friendly camera label shown in dashboard
Main RTSP URL	rtsp://admin:admin@192.168.22.225:554/...	Full-resolution stream used for AI detection
Cloudflare Live RTSP	rtsp://...sub/...	Sub-stream used for Cloudflare live view
Pi IP Address	192.168.22.101	Local IP address of the Raspberry Pi
Router IP Address	192.168.22.1	Local router IP address
Camera IP Address	192.168.22.225	Local IP address of the IP camera
Camera Config URL	https://192.168.22.225:443	URL to the camera's own web interface
Subdomain Name	RTGCAMx	Subdomain used for permanent Cloudflare live URL
Base Domain	ifill.in	Your domain — forms https://RTGCAMx.ifill.in
Tailscale Mail ID	(optional)	Tailscale VPN identity tracking identifier
MQTT Host / Port	3.111.78.82:1883	MQTT broker connection endpoint
MQTT User / Pass	—	MQTT broker credentials
MQTT Topics	RTGCAMx/command	Topics for garbage events, camera events, commands
MQTT Command Token	GCAM_SECRET_123	Auth token — all commands must include this
SFTP Host / Port	3.111.78.82:22	SFTP server for uploading images, videos, audio
SFTP Credentials	—	SFTP username and password
SFTP Paths (7 total)	/home/ftpuser/ftp/files/RTGCAMx/...	Destination paths for each file type
Pi Reboot Times	03:00:00 and 15:00:00	Daily scheduled full Pi reboot times
Router Reboot Times	03:30 and 15:30	GPIO pulse schedule for daily router reboot

4. Files and Directories

4.1 Directory Structure

Directory	Purpose
/opt/gcam/assets/	Warning audio WAV file
/opt/gcam/data/	Config and runtime state JSON files
/opt/gcam/logs/	Application log files
/opt/gcam/models/	AI detection model files
/opt/gcam/files/	Captured images
/opt/gcam/files/garbage/	Garbage detection snapshot images
/opt/gcam/files/audio/	Downloaded audio files
/opt/gcam/files/video/	Recorded event video clips
/opt/gcam/tmp/	Temporary working files
/opt/gcam/templates/	Flask web dashboard HTML templates

4.2 Key Files

File Path	Purpose
/opt/gcam/data/config.json	All configuration values (single source of truth)
/opt/gcam/data/state.json	Live runtime state updated every few seconds
/opt/gcam/data/garbage_reference.jpg	Baseline image for garbage comparison
/opt/gcam/app.py	Main application (~5,000+ lines)
/opt/gcam/controller.py	MQTT controller for service management
/opt/gcam/assets/warning.wav	Spoken warning audio file
/opt/gcam/models/MobileNetSSD_deploy.prototxt	Person/vehicle model configuration
/opt/gcam/models/MobileNetSSD_deploy.caffemodel	Person/vehicle model weights
/opt/gcam/models/garbage_yolo.onnx	Garbage object detection model (YOLO)
/usr/local/bin/read-battery-voltage.sh	INA226 battery voltage and current reader
/usr/local/bin/gcam-force-reboot.sh	Safe reboot helper script
/etc/asound.conf	ALSA USB audio device configuration
/etc/sudoers.d/gcam-reboot	Allows Pi user to reboot without password
/etc/sudoers.d/gcam-controller	Allows Pi user to manage gcam.service

5. Systemd Services and Timers

After installation the following systemd units are active and enabled to start on boot.

Unit	Type	Description
gcam.service	Service	Main G-Cam application — restarts automatically on crash
gcam-controller.service	Service	MQTT controller for Open/Close Software commands
gcam-cloudflare-ping.service	Service	Pings cloudflare.com every 15 seconds and logs result
gcam-app-restart.timer	Timer	Restarts gcam.service every 3 hours (00:00, 03:00 ... 21:00)
gcam-pi-reboot.timer	Timer	Full Pi reboot at the two daily configured times
gcam-maintenance.timer	Timer	Runs apt cleanup and log vacuum at 02:45 and 14:45

5.1 Useful Status Commands

```
# Check main app status
sudo systemctl status gcam.service --no-pager

# View live application logs
journalctl -u gcam.service -n 100 --no-pager -f

# Check controller status
sudo systemctl status gcam-controller.service --no-pager

# List all G-Cam timers
systemctl list-timers --all | grep gcam

# Check ALSA sound card
aplay -l

# Test battery voltage reader
sudo /usr/local/bin/read-battery-voltage.sh

# Detect I2C device
i2cdetect -y 1
```

6. Detection Features

6.1 Person Detection

Uses MobileNetSSD — a lightweight convolutional neural network. Detects people in the video frame and applies post-processing filters to eliminate false positives from poles, trees, and walls (based on aspect ratio, edge density, and texture analysis).

A person is confirmed only when their foot point falls inside the configured polygon geofence zone.

Person Detection — On Trigger

- Warning audio plays on the Pi speaker
- A JPEG snapshot is captured and uploaded via SFTP
- A video clip is recorded (6 seconds before + 8 seconds after the event)
- An MQTT event message is published to the camera event topic

6.2 Vehicle Detection

Uses the same MobileNetSSD model, targeting car, bus, and motorbike classes. Supports two detection modes:

Mode	Behaviour
Static Mode	Detects parked or stopped vehicles. Vehicle must remain inside the geofence for 5 continuous seconds with at least 3 confirmed detections before triggering.
Moving Mode	Additionally requires the vehicle to show motion — detected via pixel difference between consecutive frames. Suitable for entrances and driveways.

On confirmation: SFTP snapshot + video upload, combined MQTT event message published after both files are ready.

6.3 Garbage Detection

Garbage detection has two operating modes selectable from the dashboard or via MQTT command:

Mode	Behaviour
AI Mode	Takes 3 sample frames, compares each against the reference image, runs the YOLO object detector on the garbage zone, then combines change-detection bounding boxes with object detection boxes to assign a severity level.
Normal Mode	Captures a frame and uploads it without analysis. Useful for scheduled site photography where AI processing is not needed.

Garbage Severity Levels

Severity	Garbage Count	Area Change Threshold
NONE	0 objects	< 2.5% of zone area

LOW	1–2 objects	>= 2.5% of zone area
MEDIUM	3–5 objects	>= 8.0% of zone area
HIGH	6+ objects	>= 18.0% of zone area

Garbage detection runs on a configurable schedule (default: every 60 minutes) and can also be triggered manually via MQTT or the dashboard.

7. Geofence Polygon Zones

All three detection types maintain independent polygon geofences that define the active detection area inside the camera frame. Geofences are drawn interactively on the dashboard using a live snapshot as background.

Property	Detail
Coordinate format	Ratios (0.0–1.0) relative to frame width and height
Minimum points	3 vertices
Maximum points	20 vertices
Apply changes	Immediately — no service restart required
Person geofence editor	/person_geofencing
Vehicle geofence editor	/vehicle_geofencing
Garbage geofence editor	/garbage_geofencing

8. Audio System

8.1 Warning Audio

A pre-recorded spoken warning plays through the USB speaker whenever a person is detected. The warning audio system can be:

- Enabled or disabled via MQTT command or dashboard toggle
- Scheduled to a specific time window (e.g., active only 10:00–18:00)

8.2 MQTT Audio Playback

Audio files stored on the SFTP server can be played on the Pi speaker by sending an MQTT command. The audio worker downloads the file, converts it to WAV via FFmpeg if needed, then plays it through aplay.

8.3 Default Audio (Pre-loaded)

Audio files can be imported from the SFTP default audio folder to local Pi storage, then played without re-downloading. This saves bandwidth and enables offline playback.

8.4 Live Talk

The /live_talk page enables a mobile user to hold a button and speak directly through the Pi's speaker in real time. Audio is captured from the phone microphone, streamed as WebM chunks to the Pi, converted by FFmpeg, and played via aplay.

8.5 Network Audio Muting

When the internet connection drops, the system automatically mutes the speaker to 0% and clears the audio queue. When connectivity is restored, volume returns to its previous level. This prevents a queue of stale audio playing on reconnect.

9. MQTT Command Reference

All commands are sent as JSON to the configured command topic. The token field is always required.

```
{"command": "CommandName", "token": "GCAM SECRET 123"}
```

9.1 Live Stream & Camera

Command	Description
Live Stream	Creates a temporary Cloudflare public URL for the live feed (3-minute window)
Camera Live URL	Creates a temporary Cloudflare URL to the camera's own config page (controller)

9.2 Garbage Detection

Command	Description
Garbage Detect	Triggers an immediate garbage detection cycle
Garbage Sample Capture	Captures current frame as the new garbage reference image
Garbage AI Mode	Switches garbage detection to AI analysis mode
Garbage Normal Mode	Switches garbage detection to capture-only mode
Garbage Detection Set to 2hours	Changes the scheduled detection interval to 2 hours
Garbage Detection Enable	Enables garbage detection
Garbage Detection Disable	Disables garbage detection

9.3 Audio Commands

Command	Description
Audio File Play + audio_name	Downloads and plays an audio file from SFTP
Import Default Audio	Downloads all files from the SFTP default audio folder to the Pi
Erase Default Audio in Pi	Deletes all default audio files from Pi local storage
Play Default Audio - filename.mp3	Plays a file already stored on the Pi locally
Warning Audio Enable	Enables person-detected warning audio (24-hour mode)
Warning Audio Disable	Disables the warning audio
Scheduled Warning Audio + start + end	Sets a time window for warning audio playback

Scheduled Warning Disabled	Clears the schedule, warning audio active all day
Queue Clear	Stops current audio and clears all pending audio jobs
Current Pi Volume	Returns the current ALSA speaker volume percentage
Pi Volume 70%	Sets the speaker volume to specified percentage (0–100%)

9.4 Detection Controls

Command	Description
Person Detection Enable	Enables person detection
Person Detection Disable	Disables person detection
Vehicle Detection Enable	Enables vehicle detection
Vehicle Detection Disable	Disables vehicle detection
Person Video Recording Enable	Enables video clip recording on person events
Person Video Recording Disable	Disables video clip recording on person events
Vehicle Video Recording Enable	Enables video clip recording on vehicle events
Vehicle Video Recording Disable	Disables video clip recording on vehicle events

9.5 Power & Device Control

Command	Description
Power Saving Enable	Pauses camera, detection, uploads, audio, and Cloudflare tunnel
Power Saving Disable	Resumes all functions from power saving mode
Scheduled Power Saving + start + end	Sets automatic power saving hours
Device Restart	Reboots the Raspberry Pi
Close The Software	Stops gcam.service (handled by controller.py)
Open The Software	Starts gcam.service (handled by controller.py)

9.6 Diagnostics

Command	Description
Data Status	Returns full device health: network, battery, RAM, disk, version info
Software Version	Returns current version string and full feature list

10. Web Dashboard

The local web dashboard is served by Flask + Waitress on port 8080. Access it at http://<Pi_IP>:8080/ from any device on the same network.

10.1 Dashboard Pages

URL Path	Description
/	Main dashboard — all tabs including live feed, detections, controls, device health
/live	Full-screen live video only
/live_talk	Mobile-friendly push-to-talk audio page
/garbage_geofencing	Garbage polygon zone editor
/person_geofencing	Person polygon zone editor
/vehicle_geofencing	Vehicle polygon zone editor
/video_feed	Raw MJPEG video stream endpoint
/video_feed_cloudflare	Sub-stream MJPEG for Cloudflare live view

10.2 Dashboard Tabs

- Live Feed — real-time MJPEG stream with detection overlay
- Last Garbage Detection — most recent garbage analysis result and image
- Last Person Detection — most recent person event snapshot
- Last Vehicle Detection — most recent vehicle event snapshot
- Garbage Geofence — interactive polygon editor
- Person Geofence — interactive polygon editor
- Vehicle Geofence — interactive polygon editor
- Controls — enable/disable detection modes, audio, power saving
- Links & Status — Cloudflare URLs, SFTP paths, connectivity status
- Device Health — CPU, RAM, disk, battery voltage, uptime

11. Battery Monitoring

The system reads voltage and current from an INA226 power sensor connected via I2C (bus 1, address 0x40). The reader script samples 8 readings and returns averaged values as JSON.


```
{
  "battery_voltage": 12.451,
  "battery_current_a": 0.832,
  "battery_current_ma": 832.1
}
```


Status Label	Voltage Threshold	Meaning
Good	>= 12.2 V	Battery is healthy and well-charged
Medium	>= 11.7 V	Battery is at moderate charge
Low	< 11.7 V	Battery charge is low — action recommended

12. Power Saving Mode

Power saving mode can be enabled via MQTT command, dashboard toggle, or on a defined schedule. When active:

- RTSP capture stops completely
- All AI detection stops
- Live stream becomes unavailable
- Audio playback stops and the queue is cleared
- Cloudflare tunnel is closed
- Only Data Status and Power Saving commands are accepted via MQTT

The system resumes all normal functions when power saving is disabled, either manually or when the scheduled window ends.

13. Router Reboot via GPIO

The Pi can reboot the connected router by pulsing GPIO pin 17. This can be triggered automatically on schedule or via MQTT.

Step	Action
1	Set GPIO pin 17 HIGH for 1 second
2	Wait 5 seconds
3	Set GPIO pin 17 HIGH for 1 second again
4	Return pin to LOW (idle state)

The router reboot runs automatically at the two configured daily times and can also be triggered on demand via an MQTT command.

Appendix — Quick Reference

A. Battery Status Thresholds

Voltage	Status
>= 12.2 V	Good
>= 11.7 V	Medium
< 11.7 V	Low

B. Default Schedules

Schedule	Time
App restart timer	Every 3 hours (00:00, 03:00, 06:00 ... 21:00)
Pi reboot (configurable)	03:00 and 15:00 (defaults)
Router reboot (configurable)	03:30 and 15:30 (defaults)
Maintenance (apt cleanup)	02:45 and 14:45 daily
Garbage detection interval	Every 60 minutes (configurable)
Cloudflare ping	Every 15 seconds

C. Garbage Severity Quick Reference

Severity	Objects Detected	Area Change
NONE	0	< 2.5%
LOW	1–2	>= 2.5%
MEDIUM	3–5	>= 8.0%
HIGH	6+	>= 18.0%

D. Key Network Ports

Port	Purpose
8080 (TCP)	Local Flask web dashboard
1883 (TCP)	MQTT broker (default)
22 (TCP)	SFTP file transfer
554 (TCP/UDP)	RTSP video stream

443 (TCP)	Camera web config (HTTPS) / Cloudflare